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Comparative Study using Raman and Visible Spectroscopy of Cretan Extra Virgin Olive Oil Adulteration with Sunflower Oil

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ABSTRACT
The adulteration of extra virgin olive oil with low-quality and inexpensive seed oil is a serious problem in the industry. In recent years, the characterization of extra virgin olive oil adulteration with various techniques has been successfully implemented. In this work, a comparative study of Raman and visible spectroscopy is presented. These methods are rapid, noninvasive, and no sample pretreatment is required. We used both methods to study Cretan extra virgin olive oil adulterated with sunflower oil. Statistical analysis based on partial least square regression was used to determine the detection limits of the methods. Raman spectroscopy was superior in comparison to visible spectroscopy with adulteration detection limits of 3.5 and 5.5%, respectively, for the same samples. These results indicate that both techniques are suitable for olive oil quality control.

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KEYWORDS
Adulteration; extra virgin olive oil; Raman spectroscopy; ultraviolet–visible spectroscopy

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Introduction

Extra virgin olive oil (EVOO) is a natural product with superior quality compared to other edible oils. EVOO is obtained by physically pressing the olive fruit, unlike seed oil, that requires chemical treatment with solvents. EVOO is more digestible compared to seed oil and is utilized better by the human body. It is more appropriate for human nutritional needs because of the moderate levels of unsaturated fatty acids and the rich presence of vitamin E. Consequently, the commercial price of EVOO is higher compared to common vegetable oil and therefore adulteration with low-quality oil is a common practice in the industry (Boskou 1996). The authentication and control of adulteration is an important and complicated topic for the olive oil industry and several processes have been established for this purpose. Among seed oil adulterants, sunflower and corn oils are the most common because of their chemical similarity to virgin olive oil and low price (Poulli, Mousdis, and Georgious 2006). This practice of mixing virgin olive oil with low price and low-quality oils leads to consumer deception and may also cause health and safety problems (Guimet, Ferré, and Boqué 2005).

In previous years, various techniques, offering qualitative and quantitative information, have been used for the authentication and adulteration detection of olive oil.

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Chromatographic techniques such as gas chromatography–mass spectroscopy and high-performance liquid chromatography have been successfully used to classify and authenticate edible oil (EI-Hamdy and El-Fizga 1995; Christopoulou et al. 2004; Damirchi, Savage, and Dutta 2005). Additionally, nuclear magnetic resonance was also used for discrimination between edible oil, quality control, and adulteration detection in edible oil (Vlahov 1999; Vigli et al. 2003). These analytical techniques are sophisticated and time consuming, and additionally sample pretreatment is necessary.

Optical spectroscopy techniques were also used for olive oil analysis, giving high accuracy with low detection limits. Ultraviolet–visible spectroscopy was used to quantify adulterated EVOO with low-grade olive oil (Torrecilla et al. 2010). Diffuse-light absorption spectroscopy in the visible and near-infrared regions demonstrated the adulteration of EVOO by low-grade olive oil (Mignani et al. 2011). Moreover, near-infrared spectroscopy with multivariate calibration models was used to identify the adulteration of olive oil (Özdemir and Özturk 2007). Poulli et al. differentiated edible from lampante virgin olive oil and adulteration detection of virgin olive oil was accomplished by synchronous fluorescence spectroscopy (Poulli, Mousdis, and Georgiou 2005; Poulli, Mousdis, and Georgiou 2007). Guimet, Boque, and Ferre (2004) reported the clustering of commercial Spanish olive oil by excitation–emission fluorescence spectroscopy. In these studies, the fluorescence data were analyzed with chemometrics for detecting virgin olive oil adulteration with seed oil.

Vibrational spectroscopy techniques such as infrared and Raman techniques have been used for the determination of olive oil adulteration. Yang and Irudayaraj (2001) reported on the determination of EVOO adulteration with pomace oil by infrared spectroscopy. Obeidat, Khanfar, and Obeidat (2009) used infrared spectroscopy to classify edible oil, including virgin olive oil. Raman spectroscopy has also lead to promising results for the detection of olive oil adulteration. More specifically, differentiation between EVOO and hazelnut oil was achieved by Raman spectroscopy and chemometrics (López-Díez, Bianchi, and Goodacre 2003). Moreover, adulteration of olive oil with soybean, corn, rapeseed, and sunflower oils was studied by Raman spectroscopy using chemometrics (Zhang et al. 2011a, 2011b). Zou et al. (2009) reported a method for olive oil authentication, using the intensity ratio of specific Raman vibration bands.

To the best of our knowledge, in the field of olive oil authentication, only a few studies have been performed comparing two or more analytical techniques. Casale et al. (2010, 2012) reported the use of mass spectroscopy combined with near-infrared and ultraviolet–visible spectroscopy and also near-infrared, mid-infrared, and ultraviolet–visible spectroscopy for authentication and classification of olive oils. Recently, Wójcicki et al. (2015) examined the capability of spectroscopic techniques based on absorption and fluorescence in combination with statistical analysis to estimate the adulteration of EVOO with deodorized and refined olive oil.

Raman and visible spectroscopy are rapid and noninvasive analytical techniques where no sample pretreatment is required. Every material has a characteristic Raman spectrum attributed to its molecular fingerprint. Raman bands correspond to characteristic bond vibrations, hence, the Raman spectra provide details about the molecular structure of samples. The specific vibrational spectrum is an appropriate metric in the discrimination of oils. Complementarily, the bands that appear in the absorption spectra are related to the electronic excitation of characteristic families of compounds in oil. The examination of absorption spectra in the visible region can differentiate EVOO from other edible oils.

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These spectroscopic methods give the chemical fingerprint of oil and combined with multivariate analysis methods may be used for authentication and adulteration detection of oils.

In the present work, Raman (vibrational) and visible (electronic) spectroscopic techniques were used with chemometrics to characterize the spectral properties of EVOO, sunflower oil, and binary mixtures to investigate the adulteration of Greek (Cretan) EVOO with sunflower oil. Olive oil is one of the most important agricultural products of Crete. Olive oil has been consumed in Crete since the earlier Minoan period (3500 BC) and has a dominant role in the Mediterranean diet because of its high quality (Stefanoudaki, Kotsifaki, and Koutsaftakis 1999; Petrakis et al. 2008). Therefore, over the last few years, the Greek and especially Cretan olive oil export industry has become a rapidly growing sector.

A mobile Raman spectrometer, with a laser source at 786 nm, was used for this study. The mobility of the system allows for an on-site analysis of oils. The excitation source in the near-infrared decreases the possible fluorescence of chemical compounds in the samples. A significant fact was that Raman measurements were performed through the glass vial in which the oils were stored. Supplementary information about the presence of carotenoids in oils was obtained by Raman spectroscopy using a laser at 473 nm.

In parallel to Raman spectroscopy, we also used electronic absorption spectroscopy in the visible wavelength region from 400 to 800 nm using the same samples. For EVOO, the spectrum includes two areas of absorption, one typical of carotenoids and the second typical of chlorophyll pigments (Domenici et al. 2014). These two areas of absorption were absent in seed oil.

The goal and the novelty of the present work were to investigate the ability of these optical techniques to predict EVOO adulteration. The combination of spectroscopy with chemometrics is expected to allow quantitative comparison of Raman and visible spectroscopy for the characterization of olive oil.

Materials and methods

Raman spectroscopy

A mobile Raman spectrometer (HE 785, Horiba Jobin Yvon, France) was used to obtain the spectra. A semiconductor diode laser emitting at 786 nm was used as the light source. The laser source is coupled by an optical fiber to a compact probe head that contains the necessary optics for sample excitation and signal collection. A second optical fiber connects the compact probe head to a concave fixed grating spectrograph and a Peltier-cooled CCD detector (Horiba Synapse). This setup performs measurements from 100 to 3300 cm^{-1} with a spectral resolution of 4 cm^{-1}. The LabSpec software package was used for spectra acquisition and data handling. A VGA (monitoring) camera is seated inside the mobile probe head for monitoring the area under investigation. The samples were irradiated at a power of 30 mW, the acquisition time was 20 s, and the number of scans accumulated was 2. The focusing lens provides a $\times 20$ magnification.

A laboratory Raman spectrometer (Nicolet Almega XR) was also used in some measurements. A blue laser emitting at 473 nm was used with a power of 5 mW. The acquisition time was 20 s and the number of scans accumulated was 2. The laser beam was focused on the sample by a confocal microscope using a lens of $\times 10$ magnification.

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Visible spectroscopy

The absorption spectra of the oil samples were recorded on a PerkinElmer Lambda 950 spectrophotometer. The samples were placed in a quartz cuvette with a 2 mm pathlength without dilution. The absorbance was measured with a resolution of 2 nm from 400 to 800 nm.

Samples

The sunflower seed oil was purchased from a supermarket in Crete. The EVOO was obtained from a local producer in Crete. Binary mixtures of these oils were prepared with volume percentages of 2, 4, 6, 8, 10, 12.5, 15, 20, 25, 30, 40, 50, 60, 80% sunflower oil in EVOO. All samples were stored at room temperature, in the dark, and were analyzed without pretreatment.

Chemometrics

We estimated the detection limits of the optical spectroscopic techniques used in this study by applying multivariate statistical analysis based on partial least square (PLS) regression as the calibration method. Two matrices are created, one matrix X (independent variable), which contains the spectra for each sample and another matrix Y (dependent variable), which contains the actual concentration of sunflower oil in olive oil for each sample. The dimensionality of the initial independent variables was reduced to a smaller number of latent variables based on the covariance between matrices X and Y (Wold, Sjöström, and Eriksson 2001).

The predictive ability of the model was checked by cross-validation, by evaluating parameters such as the correlation coefficient R^2 and the root mean square error of cross-validation for all samples. Moreover, for evaluating the predictive ability of the model, external validation was performed by splitting the samples into calibration (10 samples) and validation/test sets (five samples). The validation set included volume percentages of 4, 10, 15, 25, and 40% sunflower oil in EVOO. In the case of external validation, the root mean square error of prediction was evaluated (Westad and Marini 2015). All the analysis was performed using Matlab R2015b with the PLS_Toolbox 8.1.

Results and discussion

The differentiation of edible oils was achieved because of the varying composition of fatty acids. In virgin olive oil, oleic acid, a monounsaturated fatty acid, is the dominant component composing 55–83% with chemical formula $CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$. The percentage of linoleic acid, a polyunsaturated fatty acid with chemical formula $CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$, is between 3.5 and 21% (Beltran et al. 2004). In contrast, seed oils (sunflower, corn, and soybean oil) contain higher concentrations of linoleic acid (Bellaloui et al. 2010; Zhang et al. 2011b).

Although the carbon chain is of the same length, linoleic acid contains one additional double bond compared to oleic acid that causes differences in the characteristic Raman band intensities. Hence, Raman spectroscopy was employed for the evaluation of olive oil quality.

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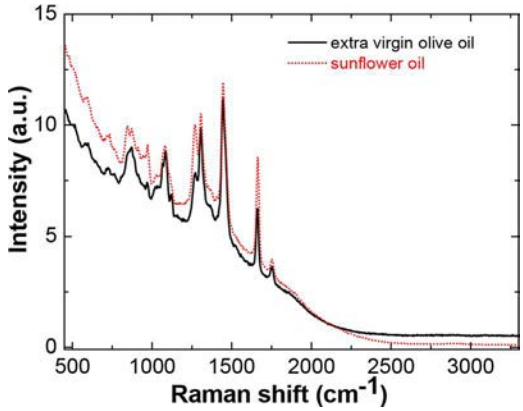

Figure 1. Raman spectra of oils from 400 to 3300 cm^{-1} without background correction. The spectrum with the black solid line corresponds to extra virgin olive oil and the spectrum with red dotted line is to sunflower oil.

Raman spectra of EVOO and sunflower oil from 400 to 3300 cm^{-1} are shown in Figure 1. Table 1 shows the characteristic Raman bands and their corresponding assignments. The original Raman spectra exhibit a high background originating from sample fluorescence. Background correction was necessary and the baseline of all Raman spectra was approximated by a third-order polynomial and subtracted from the original spectrum. Figure 2 shows Raman spectra of EVOO and sunflower oil from 1000 to 1800 cm^{-1} resulting after baseline correction. Figure 2 shows a first general discrimination of oils. In particular, the intensities of Raman bands at 1265 cm^{-1} (*cis* = C–H stretch) and 1657 cm^{-1} (*cis* C=C stretch) were higher for sunflower oil. These bands correspond to stretching vibrations of unsaturated fatty acids in oils: the band at 1265 cm^{-1} is due to monounsaturated and the one at 1657 cm^{-1} to polyunsaturated fatty acids (Baeten et al. 1998; Muik et al. 2003). These spectra are in good agreement with those reported in the literature (López-Díez, Bianchi, and Goodacre 2003; Zhang et al. 2011b).

Mixtures of extra virgin olive oil with sunflower oil

The discrimination of oil (Figure 2) was encouraging and leads to the characterization of the adulteration of EVOO with seed oil. Sunflower oil was used as the adulterant of EVOO in various volume percentages.

Table 1. Raman band assignments of the oils.

Raman band (cm^{-1})	Assignment
1082	C–C stretch, –(CH_2)–
1265	=C–H stretch, *cis* –RHC=CHR (unsaturated fatty acid)
1300	C–H bend (twist), –CH_2
1441	C=H bend, –CH_2 (saturated fatty acid)
1657	C=C stretch, *cis* –RHC=CHR (unsaturated fatty acid)
1747	C=O stretch, RC=OOR (in an ester)
2850	sym. stretch C–H stretch (CH_2) (saturated fatty acids)
2897	sym. stretch C–H stretch (CH_3)
3005	v (=C–H) stretch, *cis* RHC=CHR (unsaturated fatty acids)

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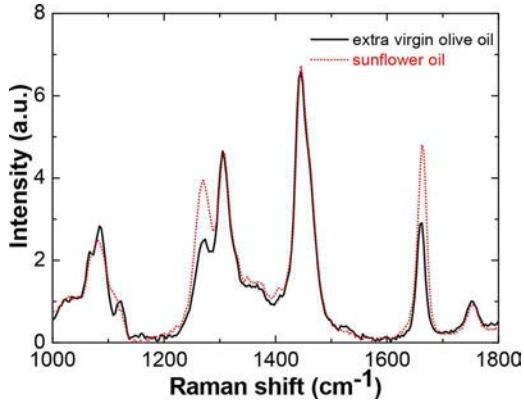

Figure 2. Raman spectra following third-order polynomial fitting subtraction from 1000 to 1800 cm^{-1}. The spectrum with the black solid line corresponds to extra virgin olive oil and the spectrum with red dotted line to sunflower oil.

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Raman spectroscopy

Figure 3 shows representative Raman spectra of mixtures of oils. Following the procedure described above, background correction of raw Raman spectra was performed. The increase in the sunflower percentage in the mixture leads to a proportional increase in Raman band intensities at 1265 and 1657 cm^{-1} corresponding to unsaturated fatty acids. Figures 2 and 3 demonstrate the differences in the intensities of specific Raman bands for EVOO, sunflower oil, and binary mixtures. These differences are derived from the different fatty acids in the samples.

We used a three latent variable PLS model of the Raman spectra with leave-one-out cross-validation for the determination of seed oil concentration in the mixture. All 15 oil samples were used to construct this model. Matrix X, described in the paragraph of chemometrics, included the spectral range from 1000 to 1800 cm^{-1}, while matrix Y

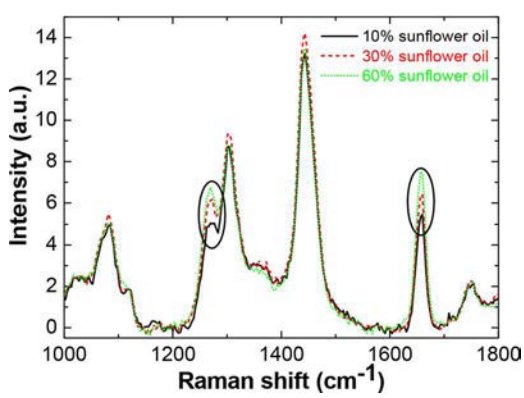

Figure 3. Raman spectra of extra virgin olive–sunflower oil mixtures from 1000 to 1800 cm^{-1}. The ellipses show characteristic Raman bands at 1265 and 1657 cm^{-1}. The spectra with black solid, red dashed, and green dotted lines correspond to 10, 30, and 60% of sunflower oil in the mixture, respectively.

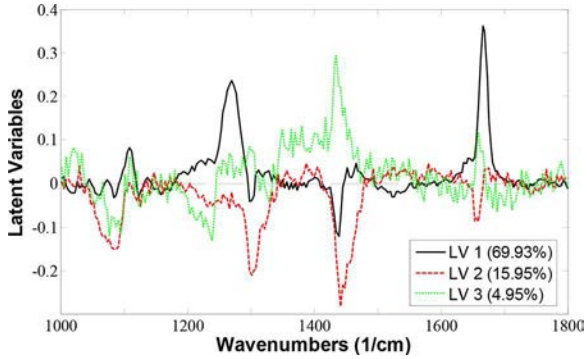

Figure 4. Partial least square loadings 1, 2, and 3 for extra virgin olive oil, sunflower oil, and mixtures. The Raman spectral range was from 1000 to 1800 cm^{-1}. First, second, and third latent variables (LV) are depicted with black solid, red dashed, and green dotted lines, respectively.

represented the actual percentage of sunflower oil adulterant. A mean-centering procedure is performed onto the data before the multivariate evaluation.

Figure 4 depicts the contribution of the three latent variables from 1000 to 1800 cm^{-1} that explained 90.83% of the spectral and 99.39% of the adulterant variation. The first latent variable explained 69.9% of the spectral and 96.8% of adulterant variation. Four main bands contributed at 1082, 1265, 1441, and 1657 cm^{-1} correspond to the C–C stretch $-(CH_2)_{n-}$, =C–H stretch (unsaturated fatty acid), C–H bend (CH_2, saturated fatty acid), and C=C stretch (unsaturated fatty acid), respectively (Table 1). These bands were related directly to the presence of saturated and unsaturated fatty acids in the oil. The correlation coefficient was 0.985 and the root mean square error of cross-validation was 3.5% by this model. The root mean square error of cross-validation is low and indicates the capability of Raman spectroscopy for measurement of EVOO adulteration with sunflower oil.

Usually EVOO adulteration with seed oils exceeds 3%. Subsequently, the 15 oil samples were divided into a calibration set containing 10 samples and a validation set consisting of five samples. The samples for validation were randomly selected covering the whole concentration range. The PLS plot in Figure 5 presents the predicted concentration of sunflower oil in the mixture based on the Raman data as a function of the actual concentration for the calibration set. The specific model had a correlation coefficient of 0.984 and root mean square error of cross-validation of 4.3%. Figure 5 also shows the predicted concentration of sunflower oil in the mixture versus the real concentration for the samples which were used as test set. The correlation coefficient was 0.988 and the root mean square error of prediction was 3.3%.

Lower detection limits have been reported for EVOO adulteration in the literature, however these studies used different seed oils, such as hazelnut and soybean oil, or different excitation sources (López-Díez, Bianchi, and Goodacre 2003; El-Abassy et al. 2009a; Mendes et al. 2015).

Raman spectroscopy with excitation at 473 nm

Raman spectroscopy employing a visible laser is used to detect carotenoids, an important family of compounds in EVOO (El-Abassy et al. 2009b). Here a laboratory Raman

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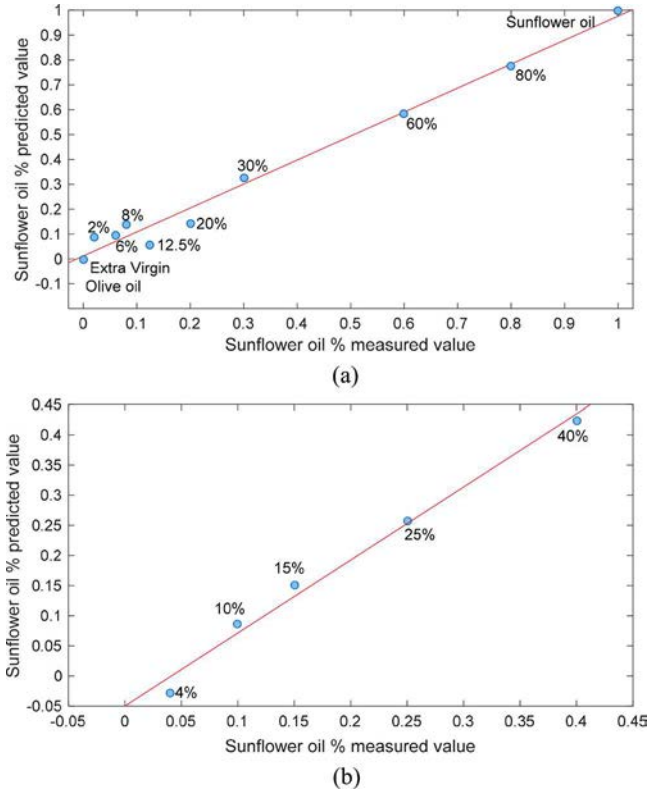

Figure 5. (a) Partial least square regression model for extra virgin olive oil, sunflower oil, and mixtures. The calibration set was composed of 10 samples. The added concentration of sunflower oil in the mixture was plotted vs. the predicted value by partial least square based on Raman data from 1000 to 1800 cm^{-1}. (b) Partial least square regression model for five samples was used as prediction set.

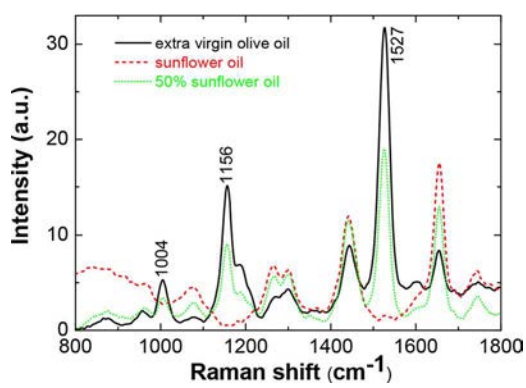

Figure 6. Raman spectra obtained with excitation at 473 nm of extra virgin olive–sunflower oil mixtures and pure samples from 800 to 1800 cm^{-1}. The bands at 1004, 1156, and 1527 cm^{-1} correspond to carotenoids. Raman spectra of extra virgin olive oil, sunflower oil, and 50% sunflower oil are denoted with black solid, red dashed, and green dotted lines, respectively.

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microscope with excitation at 473 nm was used. Figure 6 shows Raman spectra of EVOO, sunflower oil, and a mixture. In the spectra of EVOO and the mixture, three bands at 1004, 1156, and 1527 cm^{-1} were present due to carotenoids. These peaks do not appear in the Raman spectrum of pure sunflower oil. These results show that EVOO may be differentiated from sunflower oil by the characteristic vibrational Raman bands of carotenoids. The use of this source in the visible region provides additional information compared to near-infrared Raman.

Visible spectroscopy

Visible and near-infrared absorption spectroscopies have been used to differentiate olive oil, seed oil, and refined olive oil (Özdemir and Özturk 2007; Torrecilla et al. 2010; Mignani et al. 2011). Absorption spectroscopy in the visible region provides information regarding the components of oil that include carotenoids and chlorophyll. The main carotenoid pigments in olive oils are β-carotene and lutein, while pheophytin is the main chlorophyll pigment (Moyano, Heredia, and Meléndez-Martínez 2010). These peaks are absent in the visible spectrum of sunflower oil.

The absorption spectra of oil were obtained in the visible from 400 to 800 nm. The same samples were used as in the Raman analysis with volume concentrations of sunflower oil in EVOO of 2, 4, 6, 8, 10, 12.5, 15, 20, 25, 30, 40, 60, and 80%. Figure 7 shows differences in the absorption intensities of the oil samples. Pure EVOO has peaks from carotenoids at 415, 456, and 485 nm and a peak at 671 nm corresponding to chlorophyll.

Partial least square was applied to the visible spectra for the 15 oil samples. Leave-one-out cross-validation was used with the first latent variable PLS model. Matrix *X* consisted of the spectral range from 400 to 800 nm, while matrix *Y* represented the actual percentage of sunflower oil. A mean centering data pretreatment was performed before the PLS evaluation.

Figure 8 shows the contribution of the first latent variable from spectral range 400 to 800 nm, which explains 99.38% of the spectral and 97.22% of the adulterant variation. The most relevant spectral regions for predicting adulteration with sunflower oil were 400–500 nm and at 670 nm. In these regions, higher negative values for regression

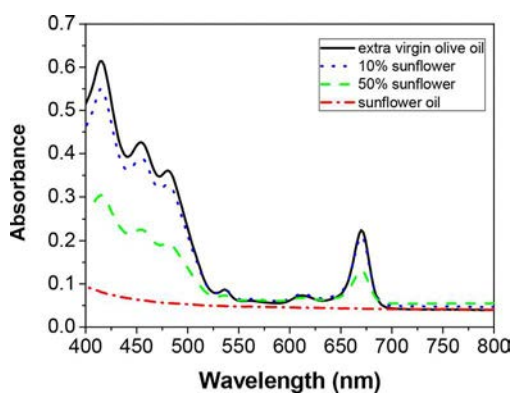

Figure 7. Absorption spectra 400–800 nm of extra virgin olive oil, sunflower oil and 50% and 10% sunflower oil in the mixture are shown with black-solid, red-dashed-dotted, green-dashed and blue-dotted line, respectively.

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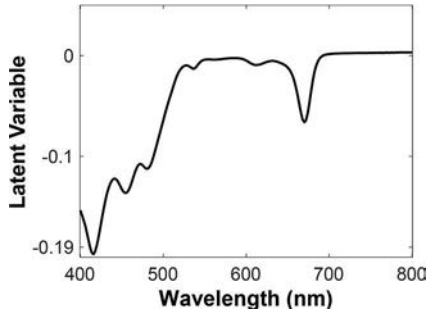

Figure 8. Partial least square loading latent variable 1 (LV1, 99.38%) for extra virgin olive oil, sunflower oil, and mixtures. The visible absorption spectral range was from 400 to 800 nm.

coefficients are obtained. The region from 400 to 500 nm relates directly to the presence of carotenoids and the region at 670 nm relates to the presence of chlorophyll in oil. EVOO and sunflower oil were differentiated by the absorption spectra in the visible region based on these compounds.

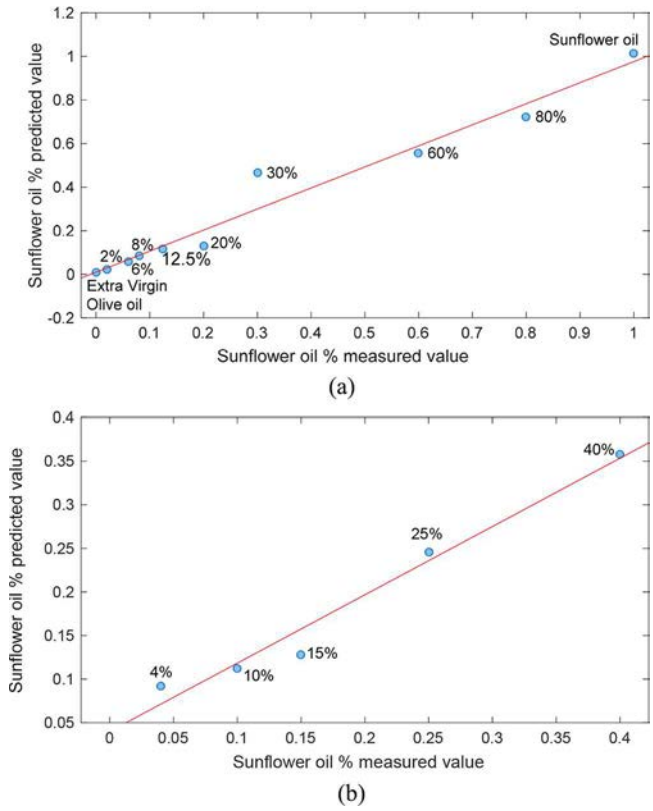

Figure 9. (a) Partial least square regression model for extra virgin olive oil, sunflower oil, and mixtures. The calibration set included 10 samples. The concentrations of sunflower oil in the mixture were plotted as a function of the predicted value from partial least squares based on visible absorption from 400 to 800 nm. (b) Partial least square regression model for the prediction set.

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The PLS model provides a correlation coefficient of 0.965 and a root mean square error of cross-validation of 5.5%. The root mean square error of cross-validation is higher than the Raman result but is within the range of commercial interest, indicating the capability of visible absorption spectroscopy for the characterization of EVOO adulteration with sunflower oil.

The samples were split into calibration (10 samples) and validation sets (5 samples). The samples for the validation were the same used in the Raman spectra. Figure 9 shows the PLS plot based on the absorption spectra of the predicted concentration of sunflower vs. the actual concentration. The PLS model shows a correlation coefficient of 0.964 and root mean square error of cross-validation of 6.4% for the calibration set. Figure 9 shows the predicted concentration of sunflower oil in the mixture as a function of the actual concentration for the test samples. The correlation coefficient was 0.971 and root mean square error of prediction was 3.2%. Downey, McIntyre, and Davies (2002) reported a lower detection limit for EVOO adulteration with sunflower oil using visible and near-infrared spectroscopy with PLS.

The Raman and visible spectroscopy results are promising for the characterization of EVOO adulteration with seed oil. The combination of spectroscopic data with PLS regression allowed quantitative measurement of adulterants in EVOO and also to a comparison of the techniques. The results for EVOO adulteration with sunflower oil for the specific study using 15 samples by Raman spectroscopy are similar to those obtained by visible absorption spectroscopy. Specifically, using a validation set of five samples, the root mean square error of prediction was 3.3 and 3.2% for Raman and visible spectroscopy, respectively.

Conclusion

Raman and visible absorption spectroscopy were used with PLS regression for the rapid characterization of EVOO adulteration with sunflower oil. The low adulteration detection limit makes these spectroscopic methods useful to the oil industry, especially for monitoring adulteration with seed oil.

The novelty of the work is based on the comparison of two techniques for olive oil adulteration which may improve methodology for characterization of EVOO treated with seed oil. The mobility of Raman spectrometer allows for an on-site analysis. Another advantage was that Raman measurements were performed through the glass vials in which the oils were stored.

This study was limited to one sample of EVOO and one sample of sunflower oil since the focus was on the comparison of the spectroscopic methods for the same samples. A quantitative comparison of these two techniques was achieved for Cretan EVOO adulterated with sunflower oil. The adulteration detection limits for the techniques were comparable, with Raman spectroscopy presenting slightly lower values.

Supplementary information about carotenoid content was obtained using a Raman system with an excitation in the visible region. The results obtained from Raman and visible absorption spectroscopy were promising for the characterization of EVOO. The next step is the investigation of a larger number of samples from various areas of Crete to construct a local olive oil database. More work is necessary for the application of this comparative methodology to other seed oils, such as corn and soybean oil, those may be used for adulterating EVOO.

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